Uniform Neumann control for strictly outward non-gradient flows

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October 12, 2026

Abstract

We prove uniform null controllability of an advection–diffusion equation with Neumann boundary conditions as the diffusion coefficient tends to zero. The drift is autonomous, belongs to $W^{2,\infty}$, points strictly outward on the boundary, and its backward flow carries a neighborhood of every point into the interior control region within a bounded time. For every sufficiently large control time, an L^2 control drives any L^2 initial state to zero with a norm bound independent of the diffusion coefficient. No gradient representation of the drift is required. A scalar function prescribed only through its boundary normal derivative leaves a tangential drift and a positive Robin coefficient in the adjoint equation. Two Carleman weights then cancel the uncontrolled tangential boundary terms. Their fixed-time observation estimate combines with the known flushing dissipation estimate to give the uniform control bound.

1 Introduction

Consider an interior control for an advection–diffusion equation on a bounded domain. Null controllability asks whether this control can drive every initial state to zero at a prescribed time. When the diffusion coefficient ε is small, the equation is close to transport, so the trajectories of the drift become relevant to the cost of control. The boundary condition is equally important: Neumann and mixed conditions can produce a cost growing exponentially in $1/\varepsilon$, even at large times [1, Theorems 2.8 and 2.9]. Thus controllability at each fixed $\varepsilon > 0$ does not by itself provide a uniform bound as $\varepsilon \to 0$.

For Neumann equations, Et-Tahri, Bárcena-Petisco, Boutaayamou and Maniar developed a Carleman calculation for tangential drifts [3, Section 6]. Their uniform controllability statement [3, Theorem 1.4] assumes backward flushing into an interior control region. A Lipschitz tangential flow preserves the boundary, however, so this hypothesis cannot hold for boundary trajectories [4, Remark 1.6]. Their subsequent work proves uniform large-time control for strictly outward gradient drifts [4, Theorem 1.5]. A gradient representation allows one to remove the first-order term by a scalar conjugation. For a general outward drift, the same direct Carleman calculation leaves boundary terms whose tangential part is not controlled by an interior observation [4, Remark 1.6]. The dissipation estimate under the trajectory condition, however, already holds for general drifts [4, Proposition 4.7(1)].

Here we retain that dissipation estimate and prove the missing fixed-time observation bound for an autonomous drift. We use a scalar conjugation to change its normal component on the boundary, while retaining its tangential component in the equation. This is sufficient for the boundary cancellation; the drift need not be a gradient.

Let $\Omega \subset \mathbb{R}^d$, $d \geq 2$, be a bounded connected smooth domain, let $\Gamma = \partial\Omega$, and let ν be the outward unit normal. Let $\varnothing \neq \omega \Subset \Omega$ be open and $B \in W^{2,\infty}(\mathbb{R}^d;\mathbb{R}^d)$. We use its $C^{1,1}$ representative and

assume

$$b := B \cdot \nu \geq c_0 > 0 \quad \text{on } \Gamma. \tag{1}$$

The flow F_s of $-B$ is globally defined. We assume that there are $T_0, r_0 > 0$ such that

$$\forall x \in \overline{\Omega} \quad \exists s_x \in (0, T_0) : \quad F_{s_x}(\mathbb{B}(x, r_0)) \subset \omega. \tag{2}$$

Here $\mathbb{B}(x, r_0)$ is the open Euclidean ball. This condition means that a whole neighborhood of each point, rather than only one trajectory, enters the control region. It is the ball version of the flushing condition used in [4], with the autonomous backward flow written explicitly.

We study

$$\begin{cases} y_t - \varepsilon \Delta y + B \cdot \nabla y = 1_\omega h & \text{in } \Omega \times (0, T), \\ \partial_\nu y = 0 & \text{on } \Gamma \times (0, T), \\ y(0) = y_0 & \text{in } \Omega. \end{cases} \tag{3}$$

The solution is understood in the variational sense specified in Section 2.

Theorem 1.1. *Under (1)–(2), there exist finite constants $T_* > 0$ and $C > 0$, depending on the fixed domain, control region and drift data, such that for every $T \geq T_*$, every $0 < \varepsilon < 1$, and every $y_0 \in L^2(\Omega)$, there is an $h \in L^2((0, T) \times \omega)$ for which the solution of (3) satisfies*

$$y(T) = 0 \quad \text{in } L^2(\Omega), \qquad \|h\|_{L^2((0,T)\times\omega)} \leq C \|y_0\|_{L^2(\Omega)}.$$

Neither T_* nor C depends on ε, and C can be chosen independently of $T \geq T_*$. No gradient condition on B is imposed.

In particular, the theorem gives the entire sufficiently-small-viscosity assertion in AIM Problem 415, with $\varepsilon_T = 1$. The time threshold obtained below is sufficient; no optimality is asserted. The result concerns an autonomous field with a strictly positive normal component. The case of a merely nonnegative normal component, and the corresponding time-dependent problem, are outside the statement.

The main choice is a solution of a Neumann Poisson problem with normal derivative $b/2$. Conjugating by its exponential leaves a drift tangent to Γ and a positive Robin coefficient. The singular potential created by the conjugation is included in the second part of the Carleman operator. Its mixed product has the same admissible scale as the first-order terms. The two weights agree on the boundary, so their tangential-gradient boundary terms cancel, and the Robin substitution leaves a positive normal term. This gives an observation bound of order $\exp(K/\varepsilon)$ on a fixed interval. Repeated flushing dissipation then dominates that exponential loss.

Section 2 constructs the conjugation. Section 3 proves fixed-time observation, and Section 4 combines it with dissipation and constructs the control. The full mixed-product calculation and the justification for L^2 terminal data are given in Appendices A and B. All norms without a specified set are spatial $L^2(\Omega)$ norms. We work over the reals; applying the argument to real and imaginary parts gives the same bound for complex data.

2 The adjoint and a boundary conjugation

Put $H = L^2(\Omega)$ and $V = H^1(\Omega)$, with H identified as a subspace of V' by its inner product. Define

$$\mathfrak{a}(p, q) = \varepsilon \int_\Omega \nabla p \cdot \nabla q + \int_\Omega (B \cdot \nabla p) q.$$

A weak solution of (3) belongs to $L^2(0,T;V)\cap H^1(0,T;V')$ and satisfies $\langle y_t,v\rangle+\mathfrak{a}(y,v)=(1_\omega h,v)$ for every $v\in V$. The form and its transpose are bounded. Moreover,

$$\mathfrak{a}(v,v)\geq\frac{\varepsilon}{2}\|\nabla v\|_2^2-\frac{\|B\|_\infty^2}{2\varepsilon}\|v\|_2^2.$$

Adding a sufficiently large multiple $\kappa(v,v)$ makes both forms coercive. Lions' variational existence theorem, in the form stated in [2, Theorem 1.1], gives existence, uniqueness and continuity in H. The shift is removed by $v(t)=e^{-\kappa t}y(t)$, with the forcing multiplied by $e^{-\kappa t}$. These are fixed-ε existence statements.

The backward adjoint with arbitrary terminal value $f\in H$ satisfies $-\langle\phi_t,v\rangle+\mathfrak{a}(v,\phi)=0$. Its strong form is

$$\begin{cases}\phi_t+\varepsilon\Delta\phi+B\cdot\nabla\phi+(\operatorname{div}B)\phi=0,\\ \varepsilon\partial_\nu\phi+b\phi=0,\\ \phi(T)=f.\end{cases}\tag{4}$$

Testing the variational equation and integrating the drift once gives

$$\frac{d}{dt}\|\phi(t)\|_2^2=2\varepsilon\|\nabla\phi(t)\|_2^2+\int_\Gamma b\phi(t)^2-\int_\Omega(\operatorname{div}B)\phi(t)^2\geq-M\|\phi(t)\|_2^2,\quad M=\|\operatorname{div}B\|_\infty.\tag{5}$$

Thus, for $t\leq r$,

$$\|\phi(t)\|_2^2\leq e^{M(r-t)}\|\phi(r)\|_2^2.\tag{6}$$

The time product rule in $L^2(V)\cap H^1(V')$ follows by time mollification; both resulting products converge in L^1. This justifies the energy identity for weak solutions.

To make the drift tangent on the boundary, set

$$k=\frac{1}{2|\Omega|}\int_\Gamma b,\qquad\Delta g=k,\qquad\partial_\nu g=b/2,\qquad\int_\Omega g=0.\tag{7}$$

The data have the exact Neumann compatibility. Since $b\in C^{1,1}(\Gamma)$, Nardi's theorem [5, Theorem 3.1], with $\alpha=1/2$, gives $g\in C^{2,1/2}(\overline{\Omega})$. Its boundary datum may be extended to the closure through a smooth collar, as in that theorem's notation.

Define

$$\begin{aligned}u&=e^{g/\varepsilon}\phi,\qquad D=B-2\nabla g,\qquad\beta=b/2,\\ W&=|\nabla g|^2-B\cdot\nabla g,\qquad c=\operatorname{div}B-k.\end{aligned}\tag{8}$$

Differentiating this multiplier in (4) yields

$$u_t+\varepsilon\Delta u+D\cdot\nabla u+(W/\varepsilon+c)u=0,\qquad\varepsilon\partial_\nu u+\beta u=0.\tag{9}$$

We have $D\cdot\nu=0$, $D,W\in W^{1,\infty}(\Omega)$, $c\in L^\infty(\Omega)$, and $\beta\geq c_0/2$. Only the boundary normal derivative of g was prescribed to cancel the normal component of D. The choice of a constant Laplacian keeps c bounded without requiring third derivatives of g.

3 Observation on a fixed interval

We first work on $Q=\Omega\times(0,1)$ and $\Sigma=\Gamma\times(0,1)$. Choose $\omega'\Subset\omega$ and $\eta\in C^2(\overline{\Omega})$ such that

$$\eta>0\text{ in }\Omega,\quad\eta=0\text{ on }\Gamma,\quad\|\eta\|_\infty=1,\quad|\nabla\eta|\geq\delta>0\text{ on }\overline{\Omega}\setminus\omega'.\tag{10}$$

The existence of this weight is the construction stated in [4, equation (64)]. Write $m = |\nabla\eta|^2$ and $a = \partial_\nu \eta$. On Γ, $\nabla\eta = a\nu$, $m = a^2$, and $a \le -\delta$.

For $\sigma \in \{+1, -1\}$, define

$$\xi_\sigma = \frac{e^{4\lambda+\sigma\lambda\eta}}{t(1-t)}, \qquad \alpha_\sigma = \frac{e^{6\lambda} - e^{4\lambda+\sigma\lambda\eta}}{t(1-t)}, \qquad w_\sigma = e^{-s\alpha_\sigma} u. \tag{11}$$

Both weights agree on Σ, and so do w_+, w_- and their time and tangential derivatives. We denote their common boundary values by ξ, α, w. The opposite spatial derivatives of the two weights will remove the tangential boundary norm.

Proposition 3.1. *There are a fixed $\lambda \ge 1$ and constants $K, C > 0$, depending only on the fixed geometry and coefficients in (9), such that every solution of (9) on Q satisfies*

$$\int_Q e^{-2s\alpha_+} \xi_+^3 u^2 \, dx\, dt \le C \int_{\omega\times(0,1)} e^{-2s\alpha_+} \xi_+^3 u^2 \, dx\, dt, \qquad s = K/\varepsilon, \quad 0 < \varepsilon < 1. \tag{12}$$

The constants are independent of ε.

Proof. We prove the estimate for the dense regular class described in Appendix B. All unlabelled integrals in this proof are over Q. For a fixed σ, omit its subscript on ξ, α, w, and put $q = D \cdot \nabla\eta$. Split the transformed equation as

$$L_1 w = -2\varepsilon s\lambda^2 m\xi w - 2\sigma\varepsilon s\lambda\xi \nabla\eta \cdot \nabla w + w_t + D \cdot \nabla w, \tag{13}$$

$$L_2 w = \varepsilon s^2\lambda^2 m\xi^2 w + \varepsilon\Delta w + s\alpha_t w - \sigma s\lambda\xi q w + (W/\varepsilon) w, \tag{14}$$

$$L_3 w = \sigma\varepsilon s\lambda\Delta\eta\, \xi w - \varepsilon s\lambda^2 m\xi w - cw. \tag{15}$$

Indeed, $\nabla\alpha = -\sigma\lambda\xi\nabla\eta$ and $\Delta\alpha = -\lambda^2 m\xi - \sigma\lambda\Delta\eta\,\xi$ imply $L_1 w + L_2 w = L_3 w$. Hence

$$\|L_1 w\|^2_{L^2(Q)} + \|L_2 w\|^2_{L^2(Q)} + 2(L_1 w, L_2 w)_{L^2(Q)} = \|L_3 w\|^2_{L^2(Q)}. \tag{16}$$

The two products generating the leading mass contribute coefficients -2 and $+3$, leaving

$$\varepsilon^2 s^3\lambda^4 \int m^2\xi^3 w^2 + \varepsilon^2 s\lambda^2 \int m\xi |\nabla w|^2, \tag{17}$$

together with a nonnegative directional-gradient square. Appendix A gives every mixed product. Before the boundary condition is imposed, their boundary contribution, apart from the potential, is

$$\begin{aligned} I_\sigma = \int_\Sigma \big[&- \sigma\varepsilon^2 s^3\lambda^3 m a \xi^3 w^2 - 2\varepsilon^2 s\lambda^2 m\xi w d_\sigma - \sigma\varepsilon^2 s\lambda a\xi d_\sigma^2 \\ &+ \sigma\varepsilon^2 s\lambda a\xi |\nabla_\Gamma w|^2 + \varepsilon(w_t + D\cdot\nabla_\Gamma w) d_\sigma - \sigma\varepsilon s^2\lambda\alpha_t\xi a w^2 \big]\, dS\, dt, \end{aligned} \tag{18}$$

where $d_\sigma = \partial_\nu w_\sigma$.

The singular potential was retained in L_2. Since W is autonomous and $D \cdot \nu = 0$, its mixed product is exactly

$$\begin{aligned} (L_1 w, (W/\varepsilon) w) = \int \big[&- s\lambda^2 W m\xi + \sigma s\lambda\xi(\nabla W\cdot\nabla\eta + W\Delta\eta) - (2\varepsilon)^{-1} \operatorname{div}(WD) \big] w^2\, dx\, dt \\ &- \sigma s\lambda \int_\Sigma W\xi a w^2\, dS\, dt. \end{aligned} \tag{19}$$

In particular, its bulk terms have absolute value at most $C\int(s\lambda^2\xi+1/\varepsilon)w^2$. They do not create a squared potential on the right of (16). The boundary terms in (19) cancel between the two signs.

The Robin condition in (9) gives

$$d_\sigma=(-\beta/\varepsilon+\sigma s\lambda\xi a)w. \tag{20}$$

Using $m=a^2$ on Γ, substitute this into (18) and sum. Terms odd in σ cancel. The scalar normal term gives $4\varepsilon s\lambda^2\beta a^2\xi w^2$; the difference of the normal squares gives $4\varepsilon s^2\lambda^2\beta a^2\xi^2 w^2$. The time term is $-2\beta w_t w$, whose integral vanishes. The remaining drift term is $-\beta D\cdot\nabla_\Gamma(w^2)$, and integration on the closed boundary gives

$$I_+ + I_- = \int_\Sigma \left[4\varepsilon s\lambda^2\beta a^2\xi + 4\varepsilon s^2\lambda^2\beta a^2\xi^2 + \operatorname{div}_\Gamma(\beta D)\right] w^2\, dS\, dt. \tag{21}$$

The divergence in this formula is bounded. Since $\beta\ge c_0/2$, $a^2\ge\delta^2$ and ξ has a positive lower bound, increasing K makes (21) nonnegative whenever $s\ge K/\varepsilon$. There is no remaining tangential-gradient boundary integral.

We give the bulk absorption explicitly. Apart from (19), the absolute values of the errors are bounded by

$$\begin{aligned} C\int[&\varepsilon^2 s^3\lambda^3\xi^3+\varepsilon^2 s^2\lambda^4\xi^2+\varepsilon s^2\lambda^3\xi^2+\varepsilon s^2\lambda^2\xi^3\\ &+s\xi^3+s\lambda\xi^2+s\lambda^2\xi]w^2+C\int[\varepsilon^2 s\lambda\xi+\varepsilon^2\lambda^2+\varepsilon]|\nabla w|^2. \end{aligned} \tag{22}$$

The terms with $s^3\lambda^3$ and $s\lambda$ have constants independent of λ and are absorbed outside ω' by first choosing λ large. Now fix λ. The estimates

$$|\xi_t|\le C\xi^2,\quad |\alpha_t|\le C\xi^2,\quad |\alpha_{tt}|\le C\xi^3,\quad |\nabla\alpha_t|\le C\lambda\xi^2$$

and the positive lower bound on ξ show that all the other ratios of (22) to (17) tend to zero as $K\to\infty$ when $s\ge K/\varepsilon$. The terms $\varepsilon s^2\xi^3$ and $s\xi^3$, for example, have ratios at most C/K and C/K^2. The ratios from (19) are at most $C/K^2+C/K^3$. Also

$$\|L_3 w\|^2_{L^2(Q)}\le C\int(\varepsilon^2 s^2\lambda^4\xi^2+\|c\|^2_\infty)w^2,$$

which is absorbed at the same scale outside ω'. The first-order cross term involving $\nabla(m\xi)$ is estimated after just one integration by parts, so only two derivatives of η are needed.

Set, with full-cylinder integrals,

$$\begin{aligned} P_\sigma &= \varepsilon^2 s^3\lambda^4\int\xi_\sigma^3 w_\sigma^2+\varepsilon^2 s\lambda^2\int\xi_\sigma|\nabla w_\sigma|^2,\\ N_\sigma &= \|L_1 w_\sigma\|^2_{L^2(Q)}+\|L_2 w_\sigma\|^2_{L^2(Q)}. \end{aligned}$$

Since $m\ge\delta^2$ outside ω', the preceding calculations imply

$$\sum_\sigma(N_\sigma+P_\sigma)\le C\sum_\sigma\left[\varepsilon^2 s^3\lambda^4\int_{\omega'\times(0,1)}\xi_\sigma^3 w_\sigma^2+\varepsilon^2 s\lambda^2\int_{\omega'\times(0,1)}\xi_\sigma|\nabla w_\sigma|^2\right]. \tag{23}$$

The added mass and gradient inside ω' are included on the right.

To remove the local gradient, recover the Laplacian from L_2:

$$H_\sigma := \varepsilon^2 \int (s\xi_\sigma)^{-1} |\Delta w_\sigma|^2 \le C(N_\sigma + P_\sigma). \tag{24}$$

Indeed, after squaring (14) and dividing by $s\xi_\sigma$, the term with $s^2\xi_\sigma^2$ is bounded by the mass in P_σ; the α_t and q terms have ratio at most C/K^2 to that mass. The potential term has ratio $C/(\varepsilon^4 s^4 \xi_\sigma^4) \le C/K^4$. Take $\theta \in C_c^\infty(\omega)$ equal to one near $\overline{\omega'}$. The interior identity

$$\int \theta^2 \xi_\sigma |\nabla w_\sigma|^2 = -\int \theta^2 \xi_\sigma w_\sigma \Delta w_\sigma - \int \nabla(\theta^2 \xi_\sigma) \cdot \nabla w_\sigma \, w_\sigma$$

and Young's inequality give, for every $\rho > 0$,

$$\varepsilon^2 s\lambda^2 \int_{\omega' \times (0,1)} \xi_\sigma |\nabla w_\sigma|^2 \le \rho H_\sigma + C_\rho \varepsilon^2 s^3 \lambda^4 \int_{\omega \times (0,1)} \xi_\sigma^3 w_\sigma^2. \tag{25}$$

Here one first absorbs half the cutoff-gradient integral and then pairs Δw_σ with w_σ using weights $(s\xi_\sigma)^{-1}$ and $(s\xi_\sigma)^3$. Choose ρ small in (25), use (24), and absorb into (23). Retain the positive mass and use $\xi_- \le \xi_+$, $\alpha_- \ge \alpha_+$ to get (12). Appendix B passes this estimate to every L^2 terminal datum. □

Proposition 3.2. *There are constants $C_o, K_o > 0$, independent of $\varepsilon \in (0,1)$, such that every adjoint (4) on a translated unit interval satisfies*

$$\|\phi(t)\|_2^2 \le C_o e^{K_o/\varepsilon} \int_t^{t+1} \|\phi(r)\|_{L^2(\omega)}^2 \, dr. \tag{26}$$

Proof. On $[1/3, 2/3] \times \overline{\Omega}$ the weight α_+ is bounded above and ξ_+ bounded below. On all of Q, the weight $e^{-2s\alpha_+}\xi_+^3$ is bounded above independently of $s \ge K \ge 1$: the exponential bounds every power of $[t(1-t)]^{-1}$. Thus Proposition 3.1 gives

$$\int_{1/3}^{2/3} \|u(t)\|_2^2 \, dt \le C e^{C/\varepsilon} \int_0^1 \|u(t)\|_{L^2(\omega)}^2 \, dt.$$

Undoing $u = e^{g/\varepsilon}\phi$ adds at most $2\,\mathrm{osc}(g)/\varepsilon$ to the exponent. Equation (6) transfers middle-time mass to the initial time. Finally, autonomy permits translation of the unit interval. □

4 Uniform observation and the control

We return to the original adjoint (4). The general-drift dissipation estimate [4, Proposition 4.7(1), equation (51)] has the following form. If $B \in L_t^\infty W_x^{1,\infty}$ has nonnegative normal component and satisfies the source's flushing condition on an ambient interval $[0, L]$, with $T_0 < L$, then

$$\|\phi(t_0 - T_0)\|_2^2 \le C_d \left[e^{-\delta_d/\varepsilon} \|\phi(t_0)\|_2^2 + \int_{t_0 - T_0}^{t_0} \|\phi(t)\|_{L^2(\omega)}^2 \, dt \right], \quad t_0 \in [T_0, L]. \tag{27}$$

This applies to weak adjoints with arbitrary L^2 terminal data. C_d is independent of ε and t_0. The decay exponent $\delta_d > 0$, specified in [4, Proposition 4.5], depends on the flushing radius, T_0 and the $L_t^\infty W_x^{1,\infty}$ drift norm. For the autonomous field this exponent can be fixed before choosing L. The prefactor C_d is allowed to depend on the subsequently fixed L.

To check the source condition, its flow $\Phi(t, t_0, x)$ solves the forward equation for B with value x at t_0. Autonomy gives

$$\Phi(t, t_0, x) = F_{t_0 - t}(x).$$

For any $x \in \overline{\Omega}$ and $t_0 \geq T_0$, choose $t = t_0 - s_x \in (t_0 - T_0, t_0)$. Equation (2) implies

$$\Phi\big(t, t_0, \overline{\mathbb{B}}(x, r_0/2)\big) \subset \omega.$$

This is its closed-ball flushing condition. The source's boundary assumption follows from (1). No gradient assumption occurs in (27).

Proof of Theorem 1.1. Choose an integer $m \geq 1$ such that $m\delta_d > K_o$, with K_o from (26), and set $L = mT_0 + 1$. Fix the dissipation prefactor on this ambient interval and increase it to $C_d \geq 1$. Iterating (27) on the m successive blocks gives

$$\|\phi(0)\|_2^2 \leq C_d^m \left[e^{-m\delta_d/\varepsilon} \|\phi(mT_0)\|_2^2 + \int_0^{mT_0} \|\phi(t)\|_{L^2(\omega)}^2 \, dt \right]. \tag{28}$$

Apply (26) on $[mT_0, L]$. Since $m\delta_d - K_o > 0$, (28) yields

$$\|\phi(0)\|_2^2 \leq C_* \int_0^L \|\phi(t)\|_{L^2(\omega)}^2 \, dt, \qquad C_* = C_d^m \max\{1, C_o\}.$$

For $T \geq L$, restrict its adjoint to $[0, L]$ with terminal datum $\phi(L)$. Uniqueness identifies this restriction with the weak adjoint just used. Consequently

$$\|\phi(0)\|_2^2 \leq C_* \int_0^T \|\phi(t)\|_{L^2(\omega)}^2 \, dt, \qquad T \geq L, \quad 0 < \varepsilon < 1. \tag{29}$$

Let $\mathcal{M}$ be the closure in $L^2((0,T) \times \omega)$ of these adjoint observations. Equation (29) makes

$$\ell(\phi|_\omega) = -(y_0, \phi(0))$$

a well-defined bounded functional, of norm at most $C_*^{1/2} \|y_0\|_2$. Extend it by continuity to $\mathcal{M}$. Its Hilbert representative $h \in \mathcal{M}$ has the same norm bound. One can construct it directly: if $\ell \neq 0$, minimize the norm on $\{z \in \mathcal{M} : \ell(z) = 1\}$. The parallelogram identity makes a minimizing sequence Cauchy. Its limit z is orthogonal to $\ker \ell$, so $h = z/\|z\|^2$ represents ℓ. If $\ell = 0$, take $h = 0$.

For the state with this control, the weak time-product rule and the two transposed forms give

$$(y(T), f) - (y_0, \phi(0)) = \int_0^T \int_\omega h\phi.$$

The right side equals $-(y_0, \phi(0))$. Therefore $(y(T), f) = 0$ for every $f \in L^2(\Omega)$, and $y(T) = 0$. Taking $T_* = L$ and $C = C_*^{1/2}$ proves all the asserted quantifiers. □

A The mixed products

Here all bulk integrals are over Q. Fix σ and use its ξ, α, w; write $v = \nabla\eta$, $m = |v|^2$, $q = D \cdot v$ and $d = \partial_\nu w$. To group the calculation, denote the four terms of L_1 in (13) by

$$A = -2\varepsilon s\lambda^2 m\xi w, \quad G = -2\sigma\varepsilon s\lambda\xi v \cdot \nabla w, \quad J = w_t, \quad Z = D \cdot \nabla w.$$

Denote the first four terms of L_2 by

$$R = \varepsilon s^2\lambda^2 m\xi^2 w, \quad H_0 = \varepsilon\Delta w, \quad S = s\alpha_t w, \quad U = -\sigma s\lambda\xi q w.$$

These letters are local to this appendix. Integration by parts gives

$$\begin{aligned}
(A,R) &= -2\varepsilon^2 s^3\lambda^4 \int m^2\xi^3 w^2,\\
(G,R) &= 3\varepsilon^2 s^3\lambda^4 \int m^2\xi^3 w^2 + \sigma\varepsilon^2 s^3\lambda^3 \int \operatorname{div}(mv)\xi^3 w^2\\
&\qquad - \sigma\varepsilon^2 s^3\lambda^3 \int_\Sigma ma\xi^3 w^2,\\
(J,R) &= -\varepsilon s^2\lambda^2 \int m\xi\xi_t w^2,\\
(Z,R) &= -\frac{\varepsilon s^2\lambda^2}{2} \int \operatorname{div}(m\xi^2 D) w^2.
\end{aligned} \tag{30}$$

In the second line one uses $\nabla\xi = \sigma\lambda\xi v$, and the drift has no boundary term because $D\cdot\nu = 0$.

The products against the Laplacian are

$$\begin{aligned}
(A,H_0) &= 2\varepsilon^2 s\lambda^2 \int m\xi|\nabla w|^2 + 2\varepsilon^2 s\lambda^2 \int \nabla(m\xi)\cdot\nabla w\, w\\
&\qquad - 2\varepsilon^2 s\lambda^2 \int_\Sigma m\xi w d,\\
(G,H_0) &= 2\sigma\varepsilon^2 s\lambda \int \xi D^2\eta(\nabla w,\nabla w) - \sigma\varepsilon^2 s\lambda \int \xi\Delta\eta|\nabla w|^2\\
&\qquad - \varepsilon^2 s\lambda^2 \int m\xi|\nabla w|^2 + 2\varepsilon^2 s\lambda^2 \int \xi(v\cdot\nabla w)^2\\
&\qquad - \sigma\varepsilon^2 s\lambda \int_\Sigma a\xi d^2 + \sigma\varepsilon^2 s\lambda \int_\Sigma a\xi|\nabla_\Gamma w|^2,\\
(J,H_0) &= \varepsilon \int_\Sigma w_t d,\\
(Z,H_0) &= \varepsilon \int_\Sigma (D\cdot\nabla_\Gamma w) d + \frac{\varepsilon}{2}\int (\operatorname{div} D)|\nabla w|^2\\
&\qquad - \varepsilon \int \sum_{i,j} (\partial_i D_j)(\partial_i w)(\partial_j w).
\end{aligned} \tag{31}$$

The cross term in the first line is bounded by

$$C\varepsilon^2\lambda^2 \int |\nabla w|^2 + C\varepsilon^2 s^2\lambda^4 \int \xi^2 w^2.$$

For the second line, apply the divergence identity to $\xi v\cdot\nabla(|\nabla w|^2)$ after the first integration by parts. This avoids differentiating η three times.

The time-weight products are

$$\begin{aligned}
(A,S) &= -2\varepsilon s^2\lambda^2 \int m\xi\alpha_t w^2,\\
(G,S) &= \sigma\varepsilon s^2\lambda \int \operatorname{div}(\xi\alpha_t v) w^2 - \sigma\varepsilon s^2\lambda \int_\Sigma \xi\alpha_t a w^2,\\
(J,S) &= -\frac{s}{2}\int \alpha_{tt} w^2,\\
(Z,S) &= -\frac{s}{2}\int \operatorname{div}(\alpha_t D) w^2.
\end{aligned} \tag{32}$$

The remaining first-order products are

$$
\begin{aligned}
(A,U) &= 2\sigma\varepsilon s^2\lambda^3 \int m\xi^2 q w^2,\\
(G,U) &= -\varepsilon s^2\lambda^2 \int \operatorname{div}(\xi^2 q v) w^2,\\
(J,U) &= \frac{\sigma s\lambda}{2} \int \xi_t q w^2,\\
(Z,U) &= \frac{\sigma s\lambda}{2} \int \operatorname{div}(\xi q D) w^2.
\end{aligned}
\tag{33}
$$

Their boundary terms vanish because $q = 0$ and $D \cdot \nu = 0$ on Γ. The last four products, against $(W/\varepsilon)w$, sum to (19). The time product is zero, while the other three are the integrations of the scalar term, of $-\sigma s\lambda W\xi v \cdot \nabla(w^2)$, and of $(2\varepsilon)^{-1} W D \cdot \nabla(w^2)$. These are all twenty mixed products. The integration order is the tangential-drift calculation of [3, Section 6], with the potential retained and the Robin substitution performed in (20)–(21). No Neumann cancellation from that source is assumed here.

B Justification for rough terminal data

We give the regularity and density argument at a fixed $\varepsilon > 0$. On a smooth bounded domain, normal collar coordinates and the fundamental theorem of calculus give, for $z \in H^1(\Omega)$ and every $\theta > 0$,

$$
\|z\|^2_{L^2(\Gamma)} \le \theta \|\nabla z\|_2^2 + C_\theta \|z\|_2^2. \tag{34}
$$

For smooth z, average its boundary squared value over a collar and estimate the integral of $2|z\partial_r z|$ by Cauchy–Schwarz and Young. The collar Jacobian is bounded above and below. Approximation in boundary graph charts passes the estimate to H^1.

For a smooth Neumann function z, integrations by parts yield

$$
\int_\Omega (\Delta z)^2 - \int_\Omega |D^2 z|^2 = \int_\Gamma \mathrm{II}(\nabla_\Gamma z, \nabla_\Gamma z), \qquad \mathrm{II}(X,Y) = (\nabla_X \nu) \cdot Y.
$$

Before using $\partial_\nu z = 0$, the boundary expression is $\Delta z \partial_\nu z - D^2 z(\nabla z, \nu)$. Differentiating the normal condition tangentially gives the displayed curvature term. Bound its absolute value by $C\|\nabla z\|^2_{L^2(\Gamma)}$ and use (34) on each derivative with θ small. This gives

$$
\|z\|_{H^2} \le C(\|\Delta z\|_2 + \|z\|_{H^1}). \tag{35}
$$

It also holds for $C^{2,1/2}$ Neumann functions. Even extension in the normal coordinate and even mollification approximate them in C^2 while preserving the normal condition; a finite partition with zero normal derivative combines the local approximations.

If $z \in H^1$ is a weak Neumann function with $\Delta z \in L^2$, put $f = -\Delta z + z$ and approximate f by smooth f_j in L^2. Nardi's theorem [5, Theorem 5.2] provides $C^{2,1/2}$ solutions z_j of $-\Delta z_j + z_j = f_j$, $\partial_\nu z_j = 0$. Their variational identity gives $\|z_j - z_l\|_{H^1} \le \|f_j - f_l\|_2$; their Laplacians are Cauchy in L^2, so (35) makes them Cauchy in H^2. The limit equals z, since testing the homogeneous $-\Delta + 1$ equation proves uniqueness. Thus (35) holds for weak Neumann functions.

For a weak Robin function v with $\Delta v \in L^2$ and $\varepsilon\partial_\nu v + bv = 0$, the multiplier $e^{2g/\varepsilon}v$ is Neumann and has an L^2 Laplacian. Its multiplier and inverse preserve H^2. Hence

$$
\|v\|_{H^2} \le C_\varepsilon(\|\Delta v\|_2 + \|v\|_{H^1}). \tag{36}
$$

This constant is used only at fixed ε.

Let $S(r)f$ denote the original adjoint after reversing time. Weak uniqueness gives its semigroup property; (6) gives $\|S(r)\| \le e^{Mr/2}$, and the continuous initial trace gives strong continuity. Choose nonnegative $\rho_n \in C_c^\infty((0,1/n))$, of integral one, and set $f_n = \int \rho_n(r)S(r)f\,dr$. Then $f_n \to f$ in L^2 and

$$\partial_q^j S(q)f_n = (-1)^j \int \rho_n^{(j)}(r)S(q+r)f\,dr. \tag{37}$$

This formula differentiates the translated compact test function, rather than presupposing differentiability of the orbit. Its initial values belong to H^1 because weak solutions are L^2 in time with values in H^1. The fixed-ε strong-solution and continuity results [4, Propositions 3.6 and 3.9] make each of these orbits continuous in H^1. The equation then makes its Laplacian continuous in L^2; (36) makes it, and each of its time derivatives, continuous in H^2. Conjugation by $e^{g/\varepsilon}$ preserves this regularity. The Sobolev Green identities and their boundary traces therefore justify every integration in Section 3 and Appendix A. The time endpoint weights vanish faster than every polynomial, so there are no temporal endpoint terms.

Finally (6) gives convergence of the approximating adjoints in $C([0,T];L^2)$ and hence in the observation norm. At each fixed ε the gauge is a bounded invertible multiplier. The weighted estimates and (26) pass to the limit with their already obtained constants. This passage is made at fixed ε and does not interchange a data-density limit with the vanishing-viscosity limit.

AI-use statement

OpenAI Codex provided substantial assistance in developing the argument, checking its calculations, locating and comparing the cited sources, writing the manuscript, and preparing the submission files. The argument and its conclusions are submitted for mathematical review.

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